CA Note

1 Basic Knowledge

Group $(G,*)$**: G1**: Closure **G2**: Associativity **G3**: Identity element **G4**: Inverse element ↓ **Division ring**: $R^\times = R\setminus\{0\}$; $R^\times$ group (under $\cdot$)

Ring $(R,+,\cdot)$**: R1**: $(R,+)$ is an abelian group **R2**: $(R,\cdot)$ is monoid. (no inverse) **R3**: Distributive law (left and right)

Ideal (I)**: I1**: $\neq\emptyset$ **I2**: $\forall a,b\in I, a-b\in I$ **I3**: $\forall r\in R, i\in I, ri, ir\in I$ **Lemma:** $I=R \Leftrightarrow 1\in I \Leftrightarrow I\cap R^\times\neq\emptyset$

Subgroup (H)**: H1**: $e\in H$ **H2**: $\forall a,b\in H, ab\in H$ **H3**: $\forall a\in H, a^{-1}\in H$ **Subring** (S)**: S1**: $1\in S$ **S2**: $\forall a,b\in S, a-b\in S$ **S3**: $\forall a,b\in S, ab\in S$

Integral Domain (ID): ID1: communicative ring **ID2**: no zero divisors **Lemma:** Finite ID ⇒ field ID ⇒ Cancellation Law

Factor Ring (R/I)**:** $a+I=b+I\Leftrightarrow a-b\in I$ $(R/I,\oplus,\odot)$ is a ring. ($1_{R/I}=1_R+I, 0_{R/I}=0_R+I$, I ideal) $\mathbb{Z}_n\cong\mathbb{Z}/n\mathbb{Z}$

Ring Homomorphism: $\varphi: R\to S$ s.t. **RH1**: $\varphi(a+b)=\varphi(a)+\varphi(b)$ **RH2**: $\varphi(ab)=\varphi(a)\varphi(b)$ **RH3**: $\varphi(1_R)=1_S$

- **1**: $\text{Ker}(\varphi)$ is an ideal. **2**: $\text{Im}(\varphi)$ is a subring. **3**: φ is 1-1 $\Leftrightarrow \text{Ker}(\varphi)=\{0_R\}$. **FIT:** $R/\text{Ker}(\varphi)\cong\text{Im}(\varphi)$ (First Isomorphism Theorem)
- **Canonical Homomorphism:** $can: R\to R/I$ s.t. $can(r)=r+I$ is a onto ring Homomorphism. $\text{Ker}(can)=I$ ↕ (∃!$\tilde\varphi$ 双射)
- **Lemma:** If $\varphi: R\to S$ ring Homomorphism s.t. $I\subseteq\text{Ker}(\varphi)$, then $\exists!\tilde\varphi: R/I\to S$ s.t. $\varphi=\tilde\varphi\circ can$ (i.e. $\tilde\varphi(r+I)=\varphi(r)$)

Lagrange's Theorem: G finite group, $H\le G$ subgroup, then $|H|\mid|G|$

2 Types of Ideals

Division: If R ring, then: $\forall f,g\in R[x]$,$_{g\text{ 首相系数可逆}}$ ⇒ $\exists q,r\in R[x]$ s.t. $f(x)=g(x)q(x)+r(x)$ and $\deg(r)<\deg(g)$ or $r=0$.

Lemma: If R ring, $g\in R[x], g\neq 0, I=(g)$. ⇒ [1] $\forall i\in R[x]/I, i=f+I, \deg(f)<\deg(g)$

g 首相系数可逆 [2] If $a,b\in R[x], \deg(a,b)<\deg(g)$ and $a+I=b+I$, then $a=b$. (系数也相同)

Remark: If $I=(g)$, g 首相系数不可逆: Find a polynomial g_1 s.t. $g_1=hg$, $h\in R[x]$ 首相系数可逆, 那么可满足 [1] 条件,[2] 只能 subset 但不能唯一表达.

Remark: 有些时候我们能用 gcd, lcm 计算 $<a,b>=<\gcd(a,b)>$ 和 $a\cap b=<\text{lcm}(a,b)>$ [If R communicative ring] 但注意! 要在能够用 Division Algorithm | 有 gcd, lcm 的环里用.

Principal Ideal: Ideal I: Iff $I=aR$ **Prime Ideal:** Ideal $I\neq R$: $ab\in I\Rightarrow a\in I$ or $b\in I$. (OR: If $a\notin I$ and $b\notin I$, then $ab\notin I$)

Maximal Ideal: Ideal $I\neq R$: If J ideal and $I\subseteq J\subseteq R$, then $J=I$ or $J=R$. ↓ I not radical $\Leftrightarrow\exists a\notin I$ s.t. $a^n\in I$ for some $n\in\mathbb{N}$

Radical|Radical Ideal: I: **Radical**: $\sqrt{I}=\{r\in R: r^n\in I \text{ for some } n\in\mathbb{N}\}$ **Radical Ideal**: If $I=\sqrt{I}$ $I\subseteq\sqrt{I}$ always.

Properties: Relationship between these kinds of ideals: **Maximal Ideals ⊆ Prime Ideals ⊆ Radical Ideals ⊆ Ideals**

1. R communicative ring, I ideal. **I prime $\Leftrightarrow$ R/I is an ID.** **I maximal $\Leftrightarrow$ R/I is a field** **I maximal ⇒ I prime**
2. R ring, $I_1,I_2,\dots,I_n\lhd R$, $I_1\cap I_2\cap\cdots\cap I_n=\{0\}$ ⇒ R isomorphic to a subring of $R/I_1\oplus R/I_2\oplus\cdots\oplus R/I_n$. $R\cong R/I_1\oplus R/I_2\oplus\cdots\oplus R/I_n$ if $I_i+I_j=R$ for $i\neq j$
3. R ring, I ideal, J ideal of R/I, $T=\{r\in R: r+I\in J\}$(reps of J). Then [1] **T: Ideal of R** [2] $I\subseteq T$ [3] $J=\{t+I: t\in T\}$ [4] $(R/I)/J\cong R/T$
   If $J=\langle j_1+I,\dots,j_n+I\rangle$ then $T=\langle j_1,\dots,j_n\rangle+I=\langle j_1,\dots,j_n,I\rangle$ Moreover, if $I=\langle i_1,\dots,i_m\rangle$, then $T=\langle j_1,\dots,j_n,i_1,\dots,i_m\rangle$
4. R communicative ring, I ideal. **I radical $\Leftrightarrow$ R/I has no nilpotent elements** except 0 **I prime ⇒ I radical** (nilpotent: $r^n=0$)
5. If R (commutative with *unity*), then any ideal $I\neq R$ is contained in a *maximal* ideal. If r is *not a unit*, then $1\notin\langle r\rangle=rR$

3 PID | UFD | Euclidean Domain (ED ⇒ PID ⇒ UFD ⇒ ID)

Irreducible: $0\neq a\in R\setminus R^\times$ is **Irreducible**: If $a=bc$, then $b\in R^\times$ or $c\in R^\times$.

Prime: $0\neq a\in R\setminus R^\times$ is **Prime**: If $a\mid bc$, then $a\mid b$ or $a\mid c$. (If R communicative ring, then prime ⇒ irreducible)

1. F field, $0\neq g\in F[x]$ non-constant: [1] **$F[x]/\langle g\rangle$ is a field** $\Leftrightarrow$ [2] **g is irreducible in $F[x]$** $\Leftrightarrow$ [3] **$F[x]/\langle g\rangle$ is a domain**
2. If R is ID: **Prime ⇒ Irreducible** If $0\neq a\in R$, then: **a prime $\Leftrightarrow$ aR prime ideal**
3. If R is PID, $a\neq 0$: [1] **aR is prime ideal** $\Leftrightarrow$ [2] **a is prime** $\Leftrightarrow$ [3] **aR is maximal ideal** ⇒ [4] **a is irreducible**

PID: R communicative ring is *principal ideal domain* if: $\forall I\lhd R, I=aR$ for some $a\in R$. Theorem: **$F, F[x]$ field is PID.**

ED: R is *ID*. R is *Euclidean Domain* if: (**Euclidean Function**) $\exists v: R\setminus\{0\}\to\mathbb{N}_0$ s.t. $\forall a,b\in R, b\neq 0, \exists q,r\in R$ s.t. $a=bq+r$ and $r=0$ or $v(r)<v(b)$.

Common ED: $\mathbb{Z}$, $F[x]$, $\mathbb{Z}[i]_{\text{Gaussian Integers}}$ are ED, with $v(a)=|a|, v(f)=\deg(f), v(a+bi)=a^2+b^2$ respectively.

Euclidean Algorithm: If R ED, $a,b\in R\setminus\{0\}$, then $\gcd(a,b)=xa+yb$ for some $x,y\in R$. (extended of Bezout's Identity)

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